

Total No. of Questions : 8]

SEAT No. :

P-9337

[Total No. of Pages : 3

[6181]-572

B.E. (Mechanical Engineering)
HYDRAULICS AND PNEUMATICS
(2015 Pattern) (Semester - I) (402041)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Figures to the right side indicate full marks.
- 2) Use of non-programmable scientific calculator is allowed.
- 3) Assume data wherever necessary and mention it.
- 4) Draw neat and suitable figure wherever necessary
- 5) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 6) Use of steam table is permitted.

- Q1)** a) State the causes of pressure losses in fluid power system. [4]
b) Why cushioning is required? In which devices it is used? State methods of cushioning. Draw labeled diagram of any one method . [8]
c) Explain with circuit "application of Pressure Reducing valve". [8]

OR

- Q2)** a) Why filters are used? Why baffles are used in reservoirs? [4]
b) Draw a labeled diagram of balanced type vane pump and explain its working. [8]
c) Explain with circuit "application of unloading valve" [8]

- Q3)** a) Explain working of Sequencing circuit A-B-A+B+ using sequencing vaves [8]
b) Explain with circuit "application of Counterbalance valve" [8]

OR

- Q4)** a) Explain synchronization circuit for 3 cylinders. State essential basic conditions for successful operation of such circuits. [8]
b) Explain with circuit "application of locking of linear actuators." [8]

P.T.O.

- Q5) a)** Explain with circuit "application of pneumatic shuttle valve." [5]
b) Explain with circuit "Pneumatic Press with safety measures". If a worker by mistake lifts his one hand while operating pneumatic press then the lift should move up. [12]

OR

- Q6) a)** Explain with circuit. "application of pneumatic twin pressure valve." [5]
b) Explain operation of "pneumatic time delay valve" with circuit [12]

- Q7)** Two identical cylinders A & B are to be operated simultaneously. The cylinder A moves again a load of 25kN, while the cylinder B moves against a load of 20kN Both the cylinders have stroke of 1m the working stroke has to be completed in 20 seconds. The return stroke of cylinder B is to start only after the cylinder A is completely retracted. The return speeds are to be fast as possible. Draw a circuit which full fill these requirements select the different components you have used in the circuit from the given data. [17]

OR

- Q8)** A pneumatic circuit is to be designed for Clamping and Drilling application. Decide the sequence and sizes of the cylinders, Clamping force required in 1kN, Drilling force required is 500N. Operating pressure is 5 bar, Assume suitable data if any. Draw circuit with labels and explain its working. Do not use sequencing valve. [17]

(a) Suction strainer:

Model	Flow Capacity (lpm)
S ₁	38
S ₂	76
S ₃	152

(b) Pressure gauge:

Model	Range (bar)
PG ₁	0 - 25
PG ₂	0 - 40
PG ₃	0 - 100
PG ₄	0 - 160

(c) Vane pump:

Model	Delivery in lpm		
	At 0 bar	At 35 bar	At 70 bar
P ₁	8.5	7.1	5.3
P ₂	12.9	11.4	9.5
P ₃	17.6	16.1	14.3
P ₄	25.1	23.8	22.4
P ₅	39.0	37.5	35.6

(d) Relief valve:

Model	Flow capacity (lpm)	Max. working pressure & bar
R ₁	11.4	70
R ₂	19.0	210
R ₃	30.4	70
R ₄	57.0	105

(e) Flow control valve:

Model	Working pressure (bar)	Flow range (lpm)
F ₁	70	0 - 4.1
F ₂	105	0 - 4.9
F ₃	105	0 - 16.3
F ₄	70	0 - 24.6

(f) Directional control valve:

Model	Max. working pressure & bar	Flow capacity (lpm)
D ₁	350	19
D ₂	210	38
D ₃	210	76

(g) Check valve:

Model	Max. working Pressure & bar	Flow capacity (lpm)
C ₁	210	15.2
C ₂	210	30.4
C ₃	210	76

(h) Pilot operated check valve:

Model	Max. working Pressure (bar)	Flow capacity (lpm)
PO ₁	210	19
PO ₂	210	38
PO ₃	210	76

(i) Cylinder (Max. working pressure 210 bar)

Model	Bore diameter (mm)	Rod diameter (mm)
A ₁	25	12.5
A ₂	40	16
A ₃	50	35
A ₄	75	45
A ₅	100	50

(j) Oil reservoirs:

Model	Capacity (litres)
T ₁	40
T ₂	100
T ₃	250
T ₄	400
T ₅	600

